

Large Language Model Based Data Augmentation for Hate Speech Detection in Roman Urdu

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Abstract

Fine grained hate speech detection in Roman Urdu remains challenging because of limited annotated data, class imbalance, informal spelling and overlapping boundaries between harmful language categories. Data augmentation offers a possible way to increase the available training data but synthetic examples may introduce incorrect labels, repetition or semantic inconsistencies if their quality is not verified. This study investigates the use of an open source Large Language Model (LLM) for quality controlled synthetic data augmentation of the RUHSOLD dataset. Mistral-7B-Instruct-v0.2 was adapted for class conditioned Roman Urdu generation and combined with a few shot prompting strategy designed to provide clearer guidance for the minority classes. A multi stage quality control pipeline was then used to assess generated samples based on basic validity, label consistency, semantic similarity to authentic data and synthetic diversity. The resulting data was evaluated using XLM-R and compared with both authentic only training and augmentation with filtered and unfiltered synthetic samples. The experiments show that adding synthetic data without quality control can reduce downstream performance while filtering substantially reduces this negative effect and can benefit individual minority classes. However, the improvements are class dependent and quality controlled augmentation does not consistently outperform training on authentic data alone. These findings highlight the importance of verifying synthetic data before using LLM generated examples for fine grained hate speech classification in low resource languages.

Index Terms—Roman Urdu, hate speech detection, large language models, synthetic data, data augmentation, natural language processing

I. INTRODUCTION

Social media platforms allow users to communicate, share opinions and engage with large online communities instantly. However, the same openness and anonymity can become the source of hateful, abusive and discriminatory content. Hate speech may target people because of characteristics such as religion, ethnicity, gender, race, nationality or political affiliation. Such content can reinforce prejudice, harming targeted individuals and communities and contribute to wider social conflict. (Fortuna and Nunes, 2018) describes hate speech as a complex social and linguistic phenomenon and emphasise on the value of tools that can identify it automatically.

Social media companies commonly rely on user reporting and human moderation to remove harmful content. These approaches are difficult to scale because online content is

produced very quickly and in many languages. Human review can be slow and subjective, particularly when moderators must interpret evolving slangs, cultural references and implicit forms of hate. Automatic hate speech detection can therefore support faster and more consistent content moderation but it does not remove the need for human oversight.

This task is much harder in the Pakistani social media context. Roman Urdu is widely used on social media where people write Urdu using Latin letters and often mix it with English. The same word can look different every time it is written because it lacks standard spelling and grammar rules. (Maab et al., 2026) point out that these spelling variations, slang words and informal structures make modeling very difficult. These challenges are compounded by the limited availability of linguistic resources, annotated datasets and language models developed specifically for Roman Urdu. There are some existing roman urdu hate speech datasets but the amount and distribution of labelled data remains limited. Datasets contain relatively small numbers of examples for some harmful categories and their class distributions are often imbalanced. Models trained on such data may learn the majority classes more effectively while failing to recognise less frequent forms of hate speech.

Research on Roman Urdu hate speech detection has progressed from traditional machine learning and recurrent neural networks to contextual transformer models but these transformer models are data hungry and requires more data. Data augmentation offers one way to address limited and imbalanced training data without collecting and annotating a new dataset from scratch. (Azam et al., 2022) evaluated several augmentation methods for Roman Urdu hate speech detection. However, traditional augmentation methods typically modify individual words or short spans of existing samples which may alter their semantic meaning or intended class label. Furthermore, the augmented samples are generally incorporated into the training data without an explicit quality check to verify their semantic consistency, label correctness and overall suitability for the intended class. This is particularly problematic for fine grained hate speech detection where small semantic changes may shift a sample from one category to another. These limitations highlight the need not only to

generate additional training data but also to assess its quality before using it for classifier training. (Jahan et al., 2024) shows that semantic drift and label alteration remain an important limitation of lexical and contextual augmentation methods.

Large Language Models (LLMs) provide a promising alternative because they can generate complete, class conditioned samples instead of applying local transformations to existing text. (Pecher et al., 2026) shows that LLM generated synthetic data can help smaller classifiers outperform direct LLM prompting in low resource multilingual classification.

This study investigates whether LLM generated synthetic data can improve fine grained Roman Urdu hate speech detection. It uses an open source instruction tuned LLM to generate additional training samples through few shot prompting. The generated data is then processed through a multi stage curation pipeline that includes basic cleaning, duplicate removal, classifier based label verification, semantic consistency filtering and synthetic diversity checking. This work further examines whether quality controlled LLM augmentation can improve overall hate speech detection performance without changing the evaluation data.

II. RELATED WORK

Automatic hate speech detection in roman urdu has received considerable attention over the past decade as social media platforms continue to face increasing volumes of harmful hate content. Early Roman Urdu hate speech detection systems primarily relied on traditional machine learning algorithms and deep neural network systems which depended heavily on handcrafted features and often struggled to capture contextual meaning. The introduction of transformer based architectures significantly improved hate speech detection by learning contextual representations of text. More recently, (Maab et al., 2026) investigated instruction tuned LLMs for Urdu and Roman Urdu offensive language detection and showed that zero shot and few shot prompting can achieve competitive performance. Despite these advances, model performance remains constrained by the limited size of available annotated datasets. One such important dataset is RUHSOLD introduced by (Rizwan et al., 2020) with both coarse grained and fine grained hate speech annotations and one of the first benchmark datasets especially for fine grained hate speech. However, its fine grained classes are highly imbalanced.

To address the scarcity and imbalance of annotated Roman Urdu hate speech data, researchers have explored several data augmentation techniques. (Azam et al., 2022) evaluated synonym replacement, random insertion, deletion, swapping, masked language model substitution and multilingual text generation on fine-grained RUHSOLD classification. Their results showed that augmentation can improve classification performance, although its effectiveness varies across augmentation methods and classifiers. Similarly, (Maqbool et al., 2024) showed that the amount of augmented data can influence downstream performance. However, these approaches largely rely on modifying or generating samples without an explicit post generation quality control pipeline. This is

particularly important for fine grained hate speech, where semantic changes may alter the intended class label.

Recent advances in Large Language Models (LLMs) have introduced a more flexible approach to data augmentation by generating entirely new training samples rather than modifying existing ones (Yoo et al., 2021). (Sahu et al., 2022) has shown that high quality synthetic data can be produced directly from off-the-shelf instruction tuned LLMs using carefully designed few shot prompts. These findings highlight the potential of LLMs as effective data generators.

Although LLM based augmentation has demonstrated promising results but very little attention has been devoted to the quality of these generated samples. (Casula and Tonelli, 2025) showed that synthetic hate speech samples should be verified before being incorporated into model training to reduce incorrectly labeled examples. More generally, (Long et al., 2024) identify sample filtering, label enhancement and quality assessment as key stages of synthetic data curation while (Jahan et al., 2024) highlight the importance of semantic consistency during augmentation. In addition, (Chim et al., 2025) argue that manual inspection should complement automatic evaluation to provide a more comprehensive assessment of synthetic data quality. However, most existing work has focused on English and other multilingual benchmarks while very little attention was paid for the quality controlled LLM augmentation of Roman Urdu hate speech detection. Motivated by this gap, this study investigates whether synthetic hate speech data generated by an open source instruction tuned LLM can improve fine grained hate speech detection when combined with a multi stage quality control pipeline before classifier training.

III. METHODOLOGY

A. Dataset Preparation

This study uses the fine grained classification task of the Roman Urdu Hate Speech and Offensive Language Detection (RUHSOLD) dataset (Rizwan et al., 2020). It was selected because it provides fine grained annotations allowing augmentation to be evaluated on specific and underrepresented forms of harmful language rather than only a binary hate/non hate distinction. The original dataset contains 10,012 Roman Urdu posts labelled as Abusive/Offensive, Normal, Religious Hate, Sexism or Profane. The three latter classes are substantially underrepresented compared with Normal and Abusive/Offensive, motivating their selection as the target classes for synthetic augmentation.

During dataset preparation, the validation file available in the RUHSOLD repository was found to be a duplicate of the training data. A replacement validation set of 801 samples was created from the released training data using stratified sampling. This produced 6,408 training samples while preserving the original 2,003 sample test set unchanged for final evaluation. Table I shows the resulting class distribution.

Preprocessing was deliberately limited to basic text cleaning and duplicate checks. Spelling normalisation, stemming and stop word removal were not applied in order to preserve the

TABLE I
CLASS DISTRIBUTION OF RUHSOLD.

Class	Train	Val.	Test
Abusive/Offensive	1537	192	481
Normal	3423	428	1070
Religious Hate	500	63	156
Sexism	537	67	168
Profane	411	51	128
Total	6408	801	2003

TABLE II
INITIAL DOWNSTREAM CLASSIFIER COMPARISON.

Model	Validation Macro-F1
mBERT	0.7068
XLM-R	0.7175
mmBERT	0.7040

spelling variation, code mixing and informal characteristics of Roman Urdu social media text.

B. Downstream Classifier Selection

Three multilingual transformer encoders were evaluated as candidate downstream classifiers: multilingual BERT (mBERT) (Devlin et al., 2019), XLM-RoBERTa-base (Conneau et al., 2020), and mmBERT-base (Marone et al., 2025). The models were trained on the same reconstructed training split and evaluated on the same validation set under a common training protocol. Macro F1 was used as the primary selection metric because the fine grained RUHSOLD classes are imbalanced and each class should contribute equally to model selection.

As shown in Table II, XLM-R achieved the highest validation macro F1 of 0.7175. Although the differences were relatively small, XLM-R provided the strongest overall balance across the five classes and was therefore selected as the downstream classifier for the subsequent augmentation experiments.

Following model selection, XLM-R hyperparameters were optimised using Optuna (Akiba et al., 2019), with validation macro F1 as the optimisation objective. The selected configuration used a learning rate of 2.8811×10^{-5} , an effective batch size of 16, weight decay of 0.03349, a warm-up ratio of 0.03756, and a maximum sequence length of 128. Training was limited to ten epochs with early stopping after two evaluations without improvement. This configuration was subsequently fixed across all final XLM-R experiments to ensure that differences between the original, unfiltered and quality controlled augmentation conditions resulted from the training data rather than changes to the classifier configuration.

C. LLM Selection and Adaptation

Mistral-7B-Instruct-v0.2 (Jiang et al., 2023) was selected as the generator based on previous evidence of its suitability for synthetic toxic content generation. Hui et al. (2025) evaluated several open source LLMs and found Mistral to be among the strongest models for harmful data synthesis. He also demonstrated that supervised fine tuning improved the reliability and diversity of generated samples compared with prompting alone. Similarly, Girón et al. (2025) selected Mistral-7B-

Instruct-v0.2 for hate speech augmentation, highlighting its balance between model capability and computational cost and its ability to generate harmful content without triggering moderation refusals observed with some alternative models. These findings provided the basis for selecting Mistral as the generator in this study.

Preliminary experiments with the base Mistral model showed that prompting alone did not reliably produce suitable Roman Urdu training samples. Although some outputs followed the requested format but generations were inconsistent in language and the alignment is weak with the intended fine grained class. This was particularly problematic because the synthetic samples needed to reflect both the linguistic characteristics of Roman Urdu and the distinctions between the RUHSOLD categories. The model was therefore adapted to the Roman Urdu domain and task using authentic RUHSOLD training examples.

Adaptation was performed using QLoRA (Dettmers et al., 2023) which combines 4-bit quantisation with Low Rank Adaptation (LoRA) (Hu et al., 2022) to reduce the computational cost of fine tuning a 7B-parameter model. Approximately 41.9 million out of 7.28 billion parameters were trainable corresponding to 0.58% of the model. Mistral was trained on authentic Roman Urdu examples together with their fine grained class information so that it could better reproduce the language and task specific characteristics of RUHSOLD. The complete training template and QLoRA configuration are provided in Appendix A-B.

D. Synthetic Data Generation Strategy

After adaptation, a series of generation experiments was conducted to determine how the fine tuned Mistral model should be prompted for synthetic data generation. Zero-shot and few-shot prompting strategies were explored using the RUHSOLD class definitions and authentic training examples. These experiments showed a clear improvement over the base model: the adapted model was substantially more capable of producing informal Roman Urdu text in the required format. However, improved language generation did not necessarily imply reliable fine grained class conditioning. The remaining challenge was therefore not primarily whether Mistral could generate Roman Urdu but whether the generated post expressed the specific type of harmful content requested by the target class.

This distinction became particularly apparent for Religious Hate and Sexism. Some generations contained religious terminology or gender associated vocabulary but did not clearly express hostility because of *religion or gender*. For example, the presence of a religious reference within a general personal or political insult was not sufficient for Religious Hate just as sexual vocabulary or references to women were not necessarily evidence of Sexism. Manual examination of the authentic RUHSOLD training examples revealed similar ambiguous cases within these classes. Representative cases are provided in Appendix A-C2.

To provide stronger guidance, class specific high precision demonstration pools were manually curated from the authentic RUHSOLD training data for the three classes targeted for augmentation. Examples were retained when manual review indicated that the defining characteristic of the class was clearly expressed: religion was the basis of hostility for Religious Hate, gender or sexual orientation was the basis of hostility or degradation for Sexism and Profane examples contained profanity without a primarily intended target. This process produced 118 Religious Hate, 40 Sexism and 40 Profane demonstrations. For each generation prompt, four unique examples were sampled from the corresponding pool so that there is a balance between class specific context and avoiding frequent reuse of the same demonstrations for the smaller Sexism and Profane pools.

The final generation prompt therefore combined three complementary forms of guidance: a definition of the requested class, an explicit description of its boundary with neighbouring categories and four authentic high precision demonstrations. The complete prompt, class specific boundary instructions and decoding configuration are reported in Appendix VII-C.

Even with clearer demonstrations and explicit boundary instructions, synthetic generation remained probabilistic and could still produce ambiguous, repetitive or incorrectly conditioned examples. The generated candidates were therefore treated as an unverified synthetic pool and were subsequently assessed using the multi-stage quality control procedure described in Section III-E.

E. Quality-Control Pipeline

An LLM-as-a-Judge approach was initially investigated as a potential semantic and label consistency filter. Prometheus 2 was selected for this experiment because it is an open source evaluator specifically trained to assess model outputs using user defined evaluation criteria and has demonstrated strong agreement with human judgements on established evaluation benchmarks (Kim et al., 2024). This made it a promising candidate for applying the task specific class definitions and quality criteria required in the present study.

To determine its capability for Roman Urdu evaluation, Prometheus 2 was evaluated against a manually labelled synthetic validation set using a five point rubric which covers Roman Urdu usability, output quality and consistency with the intended RUHSOLD class. Class consistency was the main evaluation criterion. The judge was instructed to check whether the text actually matched the intended class rather than relying only on the presence of religious, gender related or profane words. Scores of 4–5 were mapped to *Accept* while scores of 1–3 were mapped to *Reject*. The complete evaluation rubric, validation setup, results, and representative disagreements are reported in Appendix A-D.

Despite being reported as a general evaluator, Prometheus 2 showed weak agreement with the human judgements in this task specific setting. Across 79 valid evaluations, it achieved 0.519 accuracy and Cohen’s $\kappa = 0.042$. To examine whether

this behaviour was specific to the synthetic validation samples, another diagnostic was conducted using authentic RUHSOLD examples paired with both correct and incorrect class assignments. Performance remained limited with a macro-F1 of 0.550 and $\kappa = 0.160$.

Qualitative inspection further revealed errors in interpreting Roman Urdu profanity and distinguishing the fine grained class boundaries. These results indicated that the judge could not be relied upon as an automatic evaluator and therefore was excluded from the final QC pipeline rather than being retained solely on the basis of its performance on general evaluation benchmarks.

Following the Prometheus evaluation, the final QC procedure was designed as a sequence of deterministic and model based filters. Rather than relying on a single model to determine whether a synthetic example was correct, each stage addressed a different source of error: basic output validity, agreement with the intended class, semantic consistency with authentic data and redundancy within the synthetic pool.

Stage 0: Basic validity filtering.

The first stage removed outputs that were unsuitable for downstream processing before applying more computationally expensive checks. Candidates were rejected if they were empty, duplicated, outside the permitted length range or contained disallowed writing systems such as Urdu/Arabic or Devanagari script. Latin script Roman Urdu and natural English code mixing were retained because code mixing is a common characteristic of Roman Urdu social media text. This stage was intended only to remove clearly unusable generations rather than make a judgement about their semantic class.

Stage 1: Classifier based label consistency.

Candidates passing the basic checks were evaluated using the tuned XLM-R classifier described in Section III-B. A candidate was retained only when the class predicted by XLM-R matched the class requested during generation. This provided an independent label consistency signal because the generator and downstream classifier were different models. However, the classifier prediction was not treated as ground truth. Instead, it acted as a conservative screening stage before the subsequent semantic checks, since a classifier trained on the original dataset may itself reproduce ambiguities present in the training labels.

Embedding model selection.

Semantic filtering required an embedding model that could represent similar Roman Urdu texts effectively. Two multilingual sentence embedding models were compared: LaBSE (Feng et al., 2022) and multilingual-E5-large (Wang et al., 2024). Since neither model was developed specifically for Roman Urdu so their performance was compared using authentic RUHSOLD data. A balanced set of 100 samples was used as queries. FAISS (Johnson et al., 2021), an efficient similarity search library was used to compare the embedding of each query with the embeddings of the remaining training samples and retrieve its nearest neighbours. A retrieval was considered

more accurate when the retrieved examples belonged to the same fine grained class as the query.

Multilingual-E5-large achieved a higher overall Precision@5 (0.436) than LaBSE (0.374) and it performed better for the classes targeted for augmentation. It was therefore selected for the semantic similarity stages of the QC pipeline. Detailed retrieval results are provided in Appendix A-E.

Stage 2: Synthetic to real semantic consistency.

Each synthetic candidate that passed the previous stages was embedded using multilingual-E5-large and compared with authentic training examples from the same target class using FAISS. The candidate needed to be similar enough to the real examples of target class but not so similar that it was likely to be a copy or near duplicate of an existing training sample.

The similarity thresholds were determined by analysing RUHSOLD examples. To select the lower threshold, authentic validation samples were compared with training samples from the same class to examine the similarity expected between real examples. For the upper threshold, high similarity synthetic and real pairs were manually inspected to identify cases that were copies or very close paraphrases. Based on this analysis, a candidate was retained when

$$0.85 \leq s_{\text{real}} < 0.98.$$

Samples below 0.85 were rejected because they were not sufficiently similar to authentic examples from the target class while samples with similarity of 0.98 or higher were rejected as possible copies or near duplicates.

Stage 3: Synthetic diversity filtering

Passing the synthetic-to-real similarity check does not prevent the model from generating several very similar synthetic samples. Therefore, a final diversity check compared each candidate with previously accepted synthetic samples from the same class. The highest similarity found, denoted as s_{syn} , was calculated using multilingual-E5-large. A candidate was accepted only when

$$s_{\text{syn}} < 0.98.$$

Candidates with a similarity of 0.98 or higher were rejected as near duplicates. When a candidate was accepted, it was added to the synthetic reference pool and used when checking later candidates. The 0.98 threshold was taken from the near duplicate threshold established in the previous stage.

Together, the four stages were designed to address distinct failure modes rather than to treat any individual filter as a definitive measure of synthetic data quality. Stage 0 removed structurally invalid outputs, Stage 1 checked agreement with the intended label, Stage 2 assessed semantic grounding and novelty relative to authentic data and Stage 3 controlled redundancy within the growing synthetic pool. Candidates surviving all four stages formed the quality controlled synthetic dataset used in the augmentation experiments described in Section IV.

IV. EXPERIMENTAL SETUP

This section describes how the methodology presented in the previous section was applied in the final experiments. It covers the construction of the synthetic and augmented training datasets, the application of the quality control pipeline, the training of the downstream XLM-R models and the direct classification evaluation of the generation adapted Mistral model. The experimental design was structured to compare training on the original RUHSOLD data with unfiltered and QC filtered synthetic augmentation under the same evaluation setting.

A. Synthetic Dataset Construction

The synthetic dataset was constructed to evaluate whether augmenting the minority classes of RUHSOLD could improve downstream classification and whether quality control affected the usefulness of the generated samples. Augmentation was therefore applied only to Religious Hate, Sexism and Profane which were the main minority classes in the training data while Normal and Abusive/Offensive were left unchanged.

The final generation procedure described in Section III-D was used to produce raw candidates with the adapted Mistral model. The raw candidate pool contained 5,050 generations: 1,900 Religious Hate, 2,100 Sexism and 1,050 Profane candidates. A $1\times$ augmentation setting was used where the target number of accepted synthetic examples for each augmented class matched the number of authentic training examples in that class.

Two synthetic training sets were then constructed. The *QC-filtered* condition contained samples that passed all stages of the quality control pipeline described in Section III-E. From the accepted synthetic pool, 500 Religious Hate, 537 Sexism and 411 Profane examples were retained to form the final $1\times$ QC filtered set.

A matched *unfiltered* condition was constructed from the same raw generation pool without applying the proposed QC acceptance pipeline. The same class specific quantities were used: 500 Religious Hate, 537 Sexism and 411 Profane examples. Consequently, both augmented conditions contained exactly 1,448 synthetic samples and had the same final class distribution. This allowed the effect of quality control to be examined without confounding it with the amount of synthetic data.

The original RUHSOLD training set contained 6,408 examples. After augmentation, both the QC filtered and unfiltered training sets contained 7,856 examples. The validation and test sets were not augmented and synthetic examples with exact overlap with either evaluation split were excluded before classifier training.

B. Experimental Conditions

Three XLM-R experiments were used to evaluate the effect of synthetic augmentation. The first model was trained only on the authentic RUHSOLD training set and served as the baseline. The second used the authentic data together with

TABLE III
TRAINING DATA USED IN THE DOWNSTREAM XLM-R EXPERIMENTS.

Condition	Authentic	Synthetic	Total
Original	6,408	0	6,408
Unfiltered 1×	6,408	1,448	7,856
QC-filtered 1×	6,408	1,448	7,856

the matched unfiltered synthetic set, while the third used the authentic data together with the QC filtered synthetic set.

The unfiltered and QC filtered conditions contained the same number of synthetic samples for each target class to compare whether the proposed QC procedure provides value beyond simply adding more generated data.

All three XLM-R conditions used the same validation set and training configuration. Only the composition of the training data was changed. The official RUHSOLD test set was reserved for the final evaluation and was not used for model selection or hyperparameter tuning.

C. XLM-R Training Configuration

The XLM-R configuration selected during the classifier selection stage was frozen before the augmentation experiments. The same FacebookAI/xlm-roberta-base architecture and training settings were therefore used for the original, unfiltered and QC filtered conditions. Each experiment was repeated using three random seeds (42, 43, and 44) to account for training variability and assess the consistency of the observed results.

The learning rate was 2.8811×10^{-5} , with a weight decay of 0.0335 and a warm-up ratio of 0.0376. The physical training batch size was 16 with no gradient accumulation, giving an effective batch size of 16. The evaluation batch size was also 16. Input sequences were limited to 128 tokens and training was allowed to continue for a maximum of 10 epochs.

Validation macro F1 was used as the checkpoint selection metric because the fine grained task is class imbalanced and performance on the minority classes is therefore not adequately represented by accuracy alone. Early stopping with a patience of two evaluation rounds was used to reduce unnecessary training after validation performance stopped improving. The best checkpoint for each seed was retained independently.

D. Direct Mistral Classification

To provide an additional comparison, the Mistral model adapted for synthetic generation was also evaluated directly as a classifier. Importantly, this model had not been fine tuned for classification; its adaptation objective was the class conditioned Roman Urdu generation task described in Section III-C. Therefore, this experiment evaluates whether the generation adapted LLM could perform the downstream task directly through prompting.

A zero-shot classification prompt presented the definitions of all five RUHSOLD classes together with decision rules for distinguishing Religious Hate, Sexism, Abusive/Offensive, Profane and Normal. For each input post, the model was instructed to return only one class ID from 0 to 4. Generation

TABLE IV
TEST PERFORMANCE ACROSS THREE RANDOM SEEDS (MEAN $\pm$ STANDARD DEVIATION).

Training condition	Accuracy	Macro-F1
Original XLM-R	0.7943 ± 0.0058	0.7078 ± 0.0113
Unfiltered 1×	0.7815 ± 0.0118	0.6899 ± 0.0096
QC-Filtered 1×	0.7903 ± 0.0182	0.7071 ± 0.0292

was deterministic (`do_sample=False`) and limited to five new tokens. The first valid class ID in the response was extracted as the prediction.

The classification procedure was first run on the validation set to verify the prompt and output parsing procedure before final test evaluation. One of the 801 validation responses could not be parsed into a valid class ID and was recorded as a parsing failure. No additional classification fine tuning or parameter updates were performed. The complete classification prompt is provided in Appendix A-F.

V. RESULTS AND DISCUSSION

This section evaluates whether LLM generated synthetic data improves fine grained Roman Urdu hate speech classification and more specifically, whether the proposed quality control procedure affects the usefulness of that synthetic data. The analysis therefore considers three main training conditions: the original RUHSOLD training data, augmentation with unfiltered synthetic samples and augmentation with QC-filtered synthetic samples. In addition, the generation adapted Mistral model is evaluated directly as a classifier to provide the comparison requested between the generator and the downstream classification pipeline.

A. Overall Classification Performance

The overall test performance of the three XLM-R training conditions is reported in Table IV. Each experiment was repeated with three random seeds (42, 43, and 44) and the results are reported as the mean and standard deviation across these runs. Macro F1 is used as the primary evaluation metric because RUHSOLD is class imbalanced and improvements in the minority classes may not be adequately reflected by accuracy alone.

The original XLM-R baseline achieved the highest mean Macro F1 of 0.7078. Adding the unfiltered synthetic data reduced Macro F1 to 0.6899, this suggests that adding generated examples without checking their quality can introduce noise that offsets the potential benefit of increasing the minority class training data.

In contrast, the QC filtered condition achieved a mean Macro F1 of 0.7071, which was close to the original baseline. Although QC filtered augmentation did not improve the overall performance over the original training data, it performed better than unfiltered augmentation. This suggests that quality control helped remove unsuitable synthetic samples and reduced the negative effect of adding unfiltered synthetic data.

The QC filtered condition also showed greater variation across random seeds with a Macro F1 standard deviation of

TABLE V
MEAN PER-CLASS TEST F1 ACROSS THREE RANDOM SEEDS.

Class	Original	Unfiltered	QC-Filtered
Abusive/Offensive	0.6911	0.6559	0.6829
Normal	0.8990	0.8981	0.8952
Religious Hate	0.6754	0.6546	0.7028
Sexism	0.6453	0.6154	0.6286
Profane	0.6282	0.6257	0.6260

0.0292 compared with 0.0113 for the original baseline and 0.0096 for the unfiltered condition. In particular, its individual Macro F1 scores ranged from 0.6736 to 0.7276. This variation highlights the importance of evaluating augmentation across multiple training runs rather than drawing conclusions from a single seed.

B. Class-Level Analysis

The overall Macro F1 results hide important differences between individual classes. Table V therefore reports the mean per class F1 scores across the three seeds.

The clearest improvement from QC filtered augmentation occurred for Religious Hate. Its mean F1 increased from 0.6754 for the original baseline to 0.7028 after QC filtered augmentation, an absolute increase of approximately 0.027. In comparison, the unfiltered condition reduced Religious Hate F1 to 0.6546. The improvement therefore appeared only after the generated examples had passed through the QC pipeline.

The same benefit was not observed for Sexism. Its mean F1 decreased from 0.6453 for the original model to 0.6286 with QC-filtered augmentation. Nevertheless, this remained higher than the 0.6154 obtained using unfiltered synthetic data. Profane showed very little change with F1 scores of 0.6282, 0.6257, and 0.6260 for the original, unfiltered and QC-filtered conditions respectively. These results show that the effect of synthetic augmentation was class dependent rather than uniform across the three augmented categories.

A similar pattern can be seen for the classes that were not directly augmented. Abusive/Offensive F1 decreased slightly from 0.6911 to 0.6829 under QC-filtered augmentation while Normal remained relatively stable at 0.8990 and 0.8952 respectively. This suggests that adding minority class examples can also alter the decision boundaries between the augmented classes and neighbouring categories. In a fine grained task such as RUHSOLD improving recognition of one minority class may therefore involve trade-offs elsewhere in the label space.

C. Effect of Quality Controlled Augmentation

The comparison between the two augmented conditions provides a more direct assessment of the contribution of quality control. Both conditions increased the minority class training data using samples produced by the same generation approach while the QC filtered condition retained only candidates that passed the filtering procedure described in Section III-E. The lower performance of the unfiltered condition therefore shows that simply increasing the amount of synthetic training data was not sufficient.

TABLE VI
COMPARISON OF THE GENERATION ADAPTED MISTRAL CLASSIFIER WITH THE XLM-R TRAINING CONDITIONS.

Model / condition	Accuracy	Macro-F1
Mistral zero-shot	0.6331	0.4340
Original XLM-R	0.7943	0.7078
Unfiltered XLM-R	0.7815	0.6899
QC-Filtered XLM-R	0.7903	0.7071

The unfiltered model achieved a mean Macro F1 of 0.6899 compared with 0.7071 after quality control. This result supports the use of filtering before synthetic samples are incorporated into the downstream training data. However, the QC filtered model did not outperform the original baseline overall. The results therefore suggest that the main benefit of the proposed QC pipeline was not a general increase in classification performance, but the removal of synthetic examples that would otherwise have produced a larger negative effect.

This distinction is important for interpreting LLM based data augmentation. Synthetic data should not be assumed to provide useful additional supervision simply because it is fluent or generated for a specified class. As observed during the generation experiments, the fine grained RUHSOLD boundaries can remain difficult for the generator, particularly where related vocabulary occurs without the defining form of hostility. The downstream results are consistent with this observation: uncontrolled synthetic augmentation can introduce harmful training signals while filtering can reduce their effect.

D. Generator as Classifier Comparison

As an additional comparison, the generation adapted Mistral model was evaluated directly on the RUHSOLD test set using zero-shot classification. This experiment examined whether the adapted generator itself could perform the downstream classification task rather than using its outputs to augment a discriminative classifier.

Mistral achieved an accuracy of 0.6331 and a Macro F1 of 0.4340, substantially below all three XLM-R conditions. Its strongest class was Normal with an F1 of 0.8023 while performance was considerably lower for Religious Hate (0.2613), Sexism (0.3636), Profane (0.3679) and Abusive/Offensive (0.3750).

The large difference between Mistral and XLM-R should be interpreted in light of their different training objectives. Mistral was adapted for class conditioned Roman Urdu text generation and received no classification specific fine tuning. Its weaker zero-shot classification performance therefore does not imply that the adaptation was unsuccessful for its intended purpose. Instead, the result shows that the adapted generator was more useful as a source of candidate training examples than as a direct replacement for the discriminative classifier.

Overall, the experiments provide a more qualified picture of synthetic augmentation. Unfiltered LLM generated data reduced downstream performance, while quality control largely recovered this loss and improved performance for Religious Hate. However, these benefits did not extend to all augmented

classes and the QC filtered condition did not outperform the original baseline in mean Macro F1. This shows that the effectiveness of synthetic augmentation depends not only on the amount of generated data but also on its quality and the characteristics of the class being augmented.

VI. CONCLUSION AND FUTURE WORK

This study investigated the use of LLM generated synthetic data for fine grained Roman Urdu hate speech detection with particular focus on minority classes and the quality of synthetic data. Mistral-7B-Instruct-v0.2 was adapted to the RUHSOLD training data for conditional Roman Urdu generation and a few shot generation strategy was developed using manually curated high precision demonstrations. Generated candidates were then assessed using a multi stage quality control pipeline before being used to augment the minority classes of RUHSOLD.

The results show that synthetic augmentation did not provide a consistent improvement over training on authentic data alone. Adding unfiltered synthetic samples reduced downstream classification performance whereas QC filtered augmentation recovered much of this loss and produced performance close to the original XLM-R baseline. The effect was also class dependent with Religious Hate benefiting from QC filtered augmentation while similar improvements were not observed across all augmented classes. These findings indicate that synthetic data should not be assumed to provide useful additional training information solely because it is fluent or generated for a specified class. Quality and class consistency remain important, particularly for fine grained categories with closely related boundaries.

The generation experiments further showed that adapting an LLM to Roman Urdu does not by itself resolve ambiguity between fine grained hate speech categories. In particular, religious or gender related vocabulary could occur without religion or gender being the actual basis of hostility. Similar boundary cases were observed in the authentic training data making this a challenge for both generation and subsequent filtering. The results therefore highlight the importance of considering the quality of the source annotations, generation strategy and quality control procedure together when applying LLM based augmentation to subjective classification tasks.

Future work could investigate class specific augmentation strategies rather than applying the same augmentation ratio and filtering criteria across minority classes. Different classes may require different amounts of synthetic data or stronger quality control approaches including class specific threshold calibration, alternative label verification models, human-in-the-loop validation and more capable LLM based evaluators. Given the boundary ambiguity observed in RUHSOLD another important direction would be to review or re-annotate ambiguous training examples before adapting the generator and examine whether cleaner class supervision improves synthetic data quality. Finally, evaluating alternative multilingual generators and different augmentation ratios would help determine whether the findings observed with Mistral and the 1×

augmentation setting generalise to other generation models and augmentation scales.

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APPENDIX A

ADDITIONAL METHODOLOGICAL DETAILS

A. Initial Classifier Screening

Table VII reports the class-level F1 scores at the selected validation checkpoint for each candidate encoder. Checkpoints were selected using validation macro-F1, consistent with the primary metric used throughout the study.

TABLE VII
CLASS-LEVEL VALIDATION F1 SCORES FOR THE INITIAL CLASSIFIER SCREENING.

Class	mBERT	XLM-R	mmBERT
Abusive/Offensive	0.6789	0.6350	0.6685
Normal	0.8678	0.8819	0.8927
Religious Hate	0.6993	0.6929	0.6618
Sexism	0.7333	0.7111	0.7347
Profane	0.5545	0.6667	0.5625
Macro-F1	0.7068	0.7175	0.7040

B. Mistral QLoRA Adaptation

Mistral-7B-Instruct-v0.2 was adapted using supervised fine-tuning with QLoRA. Table VIII reports the selected configuration. These settings represent the configuration used in the final experiment rather than optimised hyperparameters.

TABLE VIII
QLoRA CONFIGURATION FOR MISTRAL-7B-INSTRUCT-V0.2.

Parameter	Setting
Quantisation	4-bit NF4
Double quantisation	Enabled
Compute dtype	bfloat16
LoRA rank (r)	16
LoRA alpha	32
LoRA dropout	0.05
Maximum sequence length	320
Effective batch size	8
Learning rate	2×10^{-4}
Weight decay	0.01
Optimizer	Paged AdamW 8-bit
Epochs	1
Training seed	42

LoRA was applied to the query, key, value and output attention projections, together with the gate, up and down projections. This resulted in approximately 41.9 million trainable parameters out of 7.28 billion parameters (0.58%).

1) *Fine-tuning Instruction Template*: Each authentic RUHSOLD training example was converted into an instruction–response pair. The user instruction specified the target class, its definition and the required generation format, while the corresponding authentic RUHSOLD post was used as the assistant response. The following template was used:

You are generating synthetic examples for an academic Roman Urdu hate-speech and abusive-language classification dataset.

Target class: [CLASS]

Class definition: [CLASS DEFINITION]

Generate one natural, informal Roman Urdu social media post that clearly belongs to the target class. The post itself must express the requested category rather than discuss or condemn it.

Write only in Roman Urdu using the Latin alphabet. Do not use Urdu or Arabic script. Return only the generated post without an explanation, label, quotation marks or additional text.

The [CLASS] and [CLASS DEFINITION] fields were populated according to the five RUHSOLD categories: Abusive/Offensive, Normal, Religious Hate, Sexism and Profane. The authentic post associated with the corresponding label was supplied as the target assistant response during supervised fine-tuning.

C. Synthetic Generation Strategy

This appendix provides the final prompting and decoding configuration used for synthetic-data generation. Four unique authentic demonstrations were sampled from the manually curated pool of the requested class for each generation prompt. The curated pools contained 118 Religious Hate, 40 Sexism and 40 Profane examples.

1) *Final Generation Prompt*: The following template was used for the final four-shot generation strategy. The class-specific fields and demonstrations were populated dynamically for each generated sample. The class-specific boundary rules are summarised in Table IX.

Prompt template

You are generating synthetic Roman Urdu social-media posts for an academic hate-speech classification dataset.

Target class: [CLASS]

Definition: [CLASS DEFINITION]

Important class boundary: [CLASS BOUNDARY]

Real examples from the target class:

Example 1: [DEMONSTRATION 1]

Example 2: [DEMONSTRATION 2]

Example 3: [DEMONSTRATION 3]

Example 4: [DEMONSTRATION 4]

Generate exactly one new, natural and informal Roman Urdu social-media post that clearly belongs to the target class.

Requirements:

- Write primarily in Roman Urdu using the Latin alphabet.
- Natural English code-mixing is allowed.
- [CLASS-SPECIFIC REQUIREMENTS]
- Do not copy or closely paraphrase the examples.
- Do not include a label, explanation, translation, or quotation marks.
- Return only the generated post.

TABLE IX
CLASS-SPECIFIC BOUNDARY GUIDANCE USED DURING SYNTHETIC GENERATION.

Class	Boundary guidance
Religious Hate	Religion must be the main reason for the hostility. General political criticism, personal abuse, or profanity without clear religious motivation is excluded. The generated post must express hostility because of religion, religious belief, or religious identity; merely mentioning religion is insufficient.
Sexism	Gender or sexual orientation must be the main reason for the hostility. General insults, sexual vulgarity, or references to female relatives without gender-based hatred are excluded. The generated post must express gender-based hostility, stereotyping, or degradation.
Profane	Profanity must not primarily target a specific person or group. Targeted, religion-based, and gender-based attacks belong to other classes. Profanity should instead function as a general expression, exclamation, intensifier, frustration, or vulgar remark.

2) *Representative Boundary Cases*: Representative examples illustrating the motivation for manual demonstration curation are shown in Table X. The examples contain offensive language and are reproduced solely to document methodological decisions in the generation pipeline.

D. Prometheus 2 Judge Validation

Before finalising the quality-control pipeline, an LLM-as-a-Judge approach was investigated as a possible method for verifying the quality and label consistency of generated samples. Prometheus 2

TABLE X
EXAMPLES MOTIVATING MANUAL DEMONSTRATION CURATION.

Type	Example	Reason
Authentic Sexism-labelled	aaja tjhe phir se maza dun	Sexual language is present, but gender is not clearly the basis of hostility.
Manually selected Sexism demonstration	ap ko mard banny ki acting pasand i? agr ni to hijra hi bany rahye sir ye mardo k kam hn hijro k ni.	The degradation is explicitly connected to gender identity and masculinity.

(prometheus-eval/prometheus-7b-v2.0) was evaluated because it is designed for reference-free assessment using explicit evaluation criteria and scoring rubrics. The purpose of this experiment was to determine whether the model could provide sufficiently reliable Accept/Reject decisions for fine-grained Roman Urdu synthetic data before it was used as an automatic quality-control gate.

1) *Human-Referenced Validation Setup*: A frozen validation set of 80 cases was constructed and manually assigned an *Accept* or *Reject* decision. Each case contained a Roman Urdu text, its intended RUHSOLD class and the corresponding human decision. The judge received only the intended class and text during evaluation; the human decision was withheld and used subsequently as the reference.

Prometheus assigned each candidate a score from 1 to 5. Scores of 4 or 5 were mapped to *Accept*, whereas scores of 1–3 were mapped to *Reject*. Agreement with the human reference was evaluated using accuracy, F1 for the Accept decision and Cohen’s κ .

2) *Evaluation Prompt and Rubric*: For every case, Prometheus was provided with the intended class, the candidate text and the definitions of all five RUHSOLD classes. The prompt explicitly instructed the model to evaluate the underlying meaning of the candidate rather than relying on isolated class-associated vocabulary.

The following condensed rubric reproduces the criteria used during the experiment:

Evaluation criterion: Determine whether the response is suitable for inclusion as a synthetic training example of the intended RUHSOLD class.

A suitable response should:

- be understandable Roman Urdu written primarily in the Latin alphabet;
- allow natural English code-mixing, informal spelling, abbreviations and minor grammatical variation;
- form a usable social-media-style utterance rather than model commentary, instructions, corrupted output or severely incomplete text;
- clearly satisfy the definition of the intended RUHSOLD class.

Class consistency was treated as the most important criterion. The judge was explicitly instructed not to assign a class solely because an associated word appeared in the text. For example, mentioning a religion was not sufficient for Religious Hate, while mentioning a woman or using a gendered swear word was not sufficient for Sexism.

Score 1: Clearly unsuitable; the intended class is fundamentally incorrect or the output is unusable.

Score 2: Mostly unsuitable; a major class-consistency or quality problem is present.

Score 3: Borderline or ambiguous; the intended class is not sufficiently clear.

ciently clear.

Score 4: Suitable; the intended class is clear and the sample is usable despite minor linguistic imperfections.

Score 5: Clearly suitable; the sample strongly and unambiguously satisfies the intended class.

Scores 4–5 $\rightarrow$ *Accept*; Scores 1–3 $\rightarrow$ *Reject*.

3) *Validation Results*: Prometheus produced a valid score for 79 of the 80 human-labelled cases. Its predictions showed weak agreement with the human reference, achieving an accuracy of 0.519, an Accept-class F1 of 0.424 and Cohen’s $\kappa = 0.042$. The model also showed a tendency towards rejection, predicting 53 Reject and 26 Accept decisions, compared with 39 Reject and 40 Accept decisions in the human reference.

To determine whether this behaviour was specific to the synthetic validation set, a second diagnostic was performed using authentic RUHSOLD data. Twenty-five authentic samples were selected, with five from each class. Each text was evaluated once with its correct class and once with an incorrect proposed class, producing 50 evaluator cases. Prometheus again showed limited reliability, achieving 0.580 accuracy, 0.432 Accept-class F1, 0.550 macro-F1 and Cohen’s $\kappa = 0.160$.

TABLE XI
PROMETHEUS 2 JUDGE-VALIDATION RESULTS.

Evaluation	N	Accuracy	Accept F1	κ
Human-labelled validation	79	0.519	0.424	0.042
Authentic RUHSOLD diagnostic	50	0.580	0.432	0.160

4) *Representative Disagreements*: Qualitative inspection indicated that the low agreement was not limited to borderline cases. Prometheus occasionally misinterpreted common Roman Urdu profanity and also produced decisions inconsistent with its own textual reasoning. Representative disagreements are shown in Table XII.

TABLE XII
REPRESENTATIVE DISAGREEMENTS BETWEEN PROMETHEUS 2 AND HUMAN JUDGEMENTS.

Text	Human	Judge	Observed issue
oye bhenchod	Accept	Reject	The judge treated the expression as potentially non-offensive despite the presence of common Roman Urdu profanity.
acha hai tu phir tere baare mei sharam nai hai	Reject	Accept	The feedback stated that the text did not clearly express gender-based hostility, but the model nevertheless returned an Accept score.
apnay baap ko sikhao to hijra	Accept	Reject	The judge recognised the derogatory content but rejected its consistency with the intended Sexism label.

Overall, the two validation experiments indicated that Prometheus 2 was not sufficiently reliable for automatic Roman Urdu fine-grained label verification. It was therefore excluded from the final quality-control pipeline rather than being used as an acceptance gate.

E. Embedding Model Selection

The semantic similarity stage required an embedding model capable of representing informal Roman Urdu text. LaBSE (Feng et al., 2022) and multilingual-E5-large (Wang et al., 2024) were therefore compared using authentic RUHSOLD examples.

A balanced set of 100 authentic samples was used as queries. The remaining training examples formed the retrieval pool. For each query, FAISS (Johnson et al., 2021) was used to retrieve the nearest neighbours based on cosine similarity. Retrieval performance was measured using Precision@5 and Precision@10, where a retrieved example was considered relevant when it had the same fine-grained class label as the query.

Table XIII summarises the overall retrieval performance.

TABLE XIII
COMPARISON OF SENTENCE EMBEDDING MODELS ON AUTHENTIC
RUHSOLD RETRIEVAL.

Embedding Model	P@5	P@10
LaBSE	0.374	0.367
multilingual-E5-large	0.436	0.405

Multilingual-E5-large achieved higher overall retrieval precision and also performed better for the three classes targeted for augmentation: Religious Hate, Sexism and Profane. It was therefore selected for the semantic similarity stages of the final QC pipeline.

F. Mistral Zero-Shot Classification Prompt

The following zero-shot prompt was used to evaluate the generation-adapted Mistral model directly on the five-class RUHSOLD classification task. No classification-specific fine-tuning was performed.

You are a classifier for the RUHSOLD Roman Urdu hate speech dataset.

Classify the given Roman Urdu social-media post into exactly ONE of the following five classes.

0 = Abusive/Offensive

Insults, personal attacks, humiliation, or offensive language that is not primarily based on religion or gender and is not primarily simple profanity.

1 = Normal

Neutral or ordinary non-abusive language without hate speech, targeted abuse, or profanity.

2 = Religious Hate

Hateful, abusive, insulting, or discriminatory language primarily targeting a religion, religious group, sect, religious identity, or religious affiliation.

3 = Sexism

Hateful, abusive, insulting, discriminatory, or derogatory language primarily targeting a person or group based on gender, or containing gender-based degradation.

4 = Profane

Language whose main characteristic is profanity, vulgarity, obscene language, or swearing, without the primary target being religion or gender.

Important decision rule:

If the post primarily targets religion, choose Religious Hate.

If the post primarily targets gender, choose Sexism.

If it is mainly general abuse or a personal attack, choose Abusive/Offensive.

If it is mainly profanity without a more specific religious or gender target, choose Profane.

If it is non-abusive, choose Normal.

Return ONLY the class ID: 0, 1, 2, 3, or 4.

Do not explain your answer.

Input:

Post: [ROMAN URDU POST]